

Non-Smooth Equilibria in Hill-Type Muscle Models

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poster

1 Abstract

Hill-type muscle models with switched activation dynamics have four directional linearizations at equilibrium. In a simulated elbow, choosing the wrong branch causes 14–75% prediction error. In closed-loop MPC, relinearizing at the measured state reduces lowering overshoot from 0.37° to 0.02°, showing that branch selection matters for prediction and transient control.

2 Introduction

Muscle models connect excitation to force in prostheses, exoskeletons, functional electrical stimulation (FES) and tendon-driven robots. LQR and successive-linearization MPC use a local Jacobian to predict motion [1,2].

But an equilibrium lies on two switches: excitation equals activation, and fibre velocity is zero. The perturbation direction selects one of four local models. **We quantify the prediction error and test how much survives feedback.**

Figure 1: Project overview

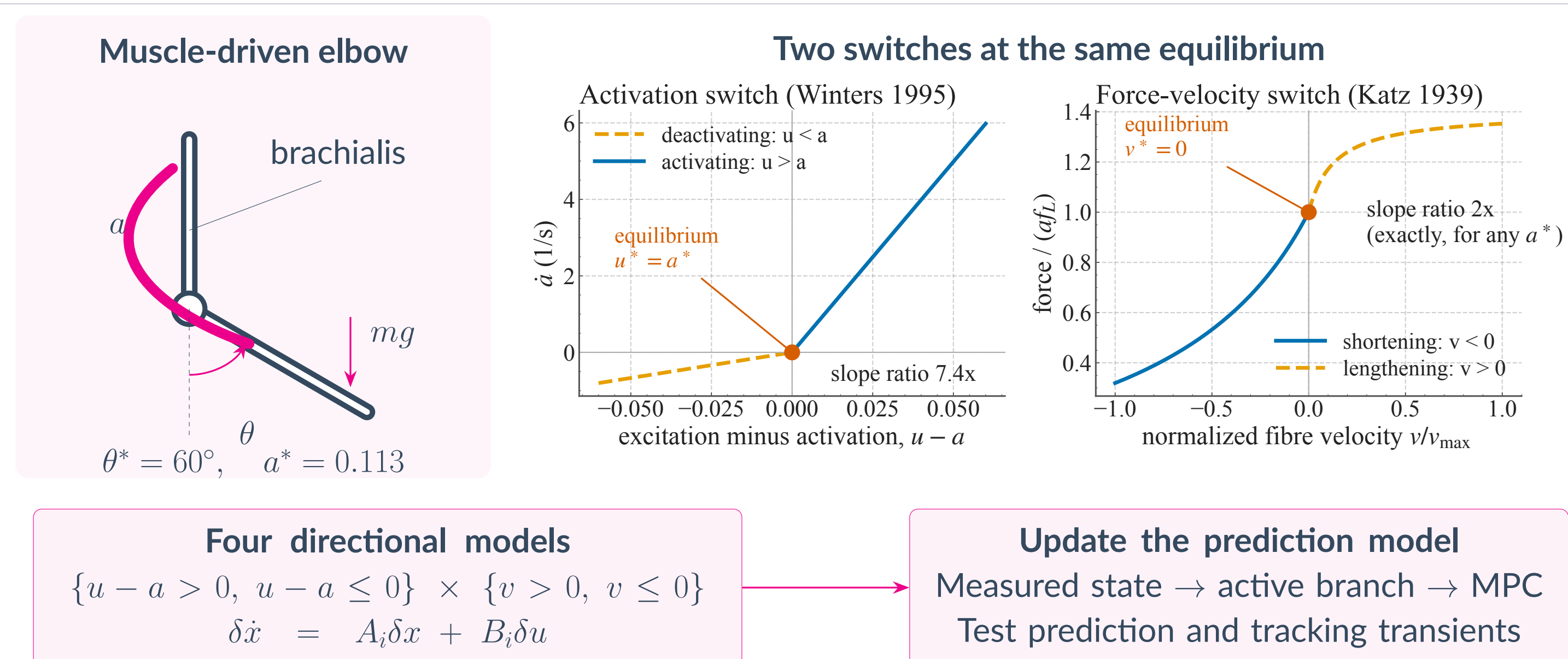


Figure 1. Project overview: a muscle-driven joint sits on two switching surfaces. The perturbation selects one of four local models; branch-aware MPC updates its prediction at the measured state. Elbow drawing is schematic.

3 Related work

Hill-type modelling. Thelen models activation and force generation; Millard et al. characterize computational muscle formulations [1,2].

Smoothing. Kirsch et al. smooth torque-velocity dynamics; De Groote et al. smooth activation for optimization [3,4]. Here we measure the cost of branch choice at equilibrium.

Non-smooth systems. Camlibel et al. give continuity conditions for bimodal linear systems [5]. Yeo et al. study a distinct force-length instability, rather than the activation-switch control problem [6].

4 Formulation / model

A forearm-hand pendulum is driven by a rigid-tendon Hill-type brachialis model: Thelen activation dynamics [1], Holzbaur muscle parameters [7], and moment arms digitized from Murray et al. [8].

$$x = [\theta, \dot{\theta}, a]^T, \quad I\ddot{\theta} = F^{MT}r(\theta) - mgl_c \sin \theta, \quad v = -\frac{r(\theta)\dot{\theta}}{l_0^M}.$$

$$\dot{a} = \frac{u - a}{\tau_a}, \quad \tau_a = \begin{cases} \tau_{\text{act}}(0.5 + 1.5a), & u > a, \\ \tau_{\text{deact}}/(0.5 + 1.5a), & u \leq a. \end{cases}$$

At $\theta^* = 60^\circ$ and $a^* = 0.113$, all four branches are stable. Each switch satisfies $A_1 - A_2 = ec^T$ and $B_1 - B_2 = ed$ [5]; the switches affect different Jacobian entries.

Strength check: the three elbow flexors peak at 73.6 N m at 94°, against 79 N m at 100° in one measured data set (RMS difference 3.6 N m). This is a simulation study; hardware validation remains open.

9 Conclusion

Choose the branch as well as the operating point. Directional linearizations matter for prediction and transient control. Relinearize at the measured state, avoid central differences across a kink, and compensate delay. Co-contraction can reduce the velocity kink.

Limits: one simulated joint; no reflexes, short-range stiffness or history dependence. Tracking rankings depend on controller weights and noise.

References

- [1] Thelen DG (2003). *J Biomech Eng* 125:70–77.
- [2] Millard M et al. (2013). *J Biomech Eng* 135:021005.
- [3] Kirsch NA et al. (2018). *IEEE TNSRE* 26:224–232.
- [4] De Groote F et al. (2016). *Ann Biomed Eng* 44:2922–2936.
- [5] Camlibel MK et al. (2008). *Automatica* 44:1261–1267.
- [6] Yeo SH et al. (2023). *J R Soc Interface* 20:20220430.
- [7] Holzbaur KRS et al. (2005). *Ann Biomed Eng* 33:829–840.
- [8] Murray WM et al. (1995). *J Biomech* 28:513–525.

5 Results: prediction error

A +0.01 excitation step at the reference posture separates the four local predictions. At 50 ms, the matched model errs by **1.1%** of the true swing; the wrong activation, velocity and combined branches err by **71%, 14% and 75%**, respectively.

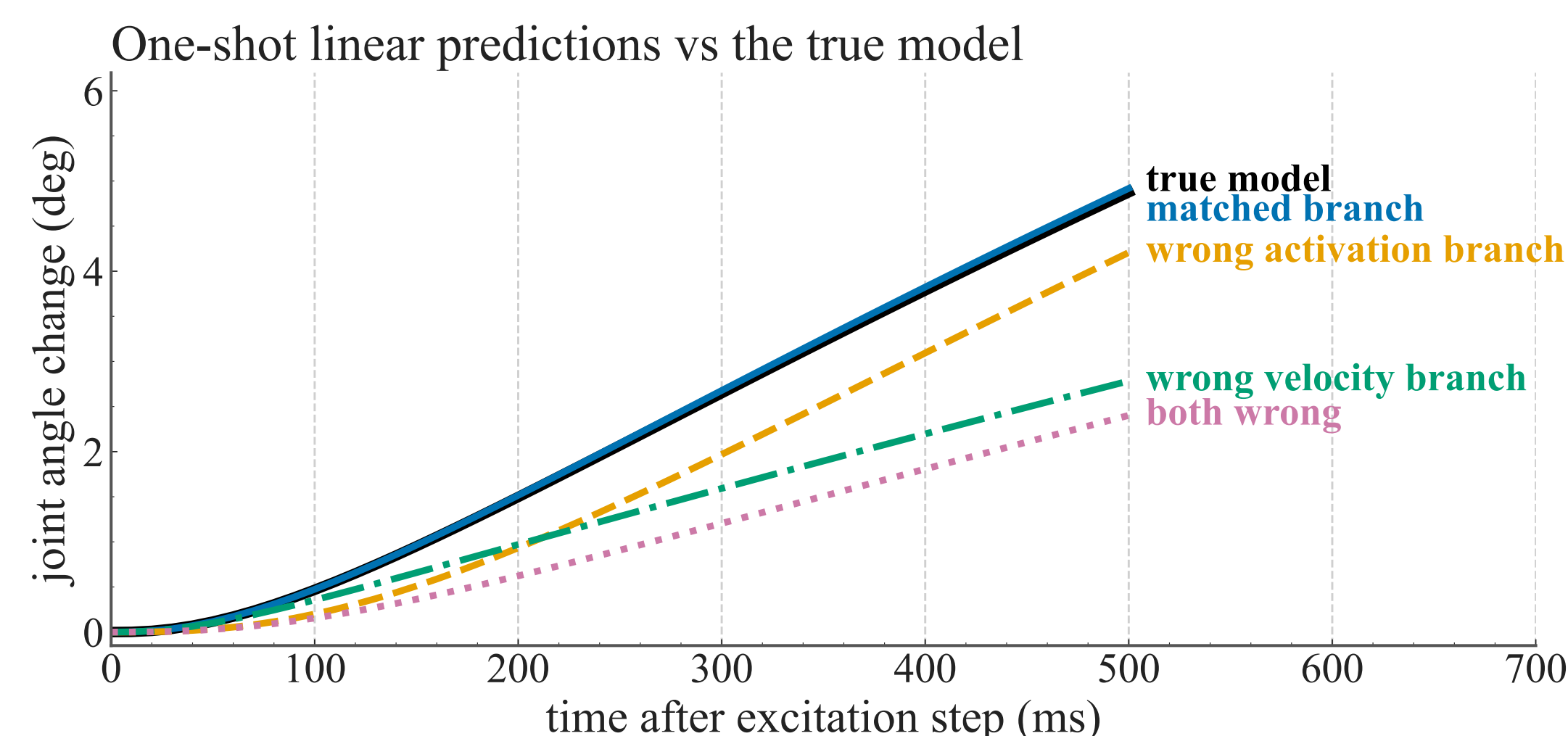


Figure 2. Only the matched branch closely tracks the nonlinear response.

6 Results: across operating points

With both branches wrong, error is at least **23%** of the activating-step signal at every single-muscle equilibrium tested.

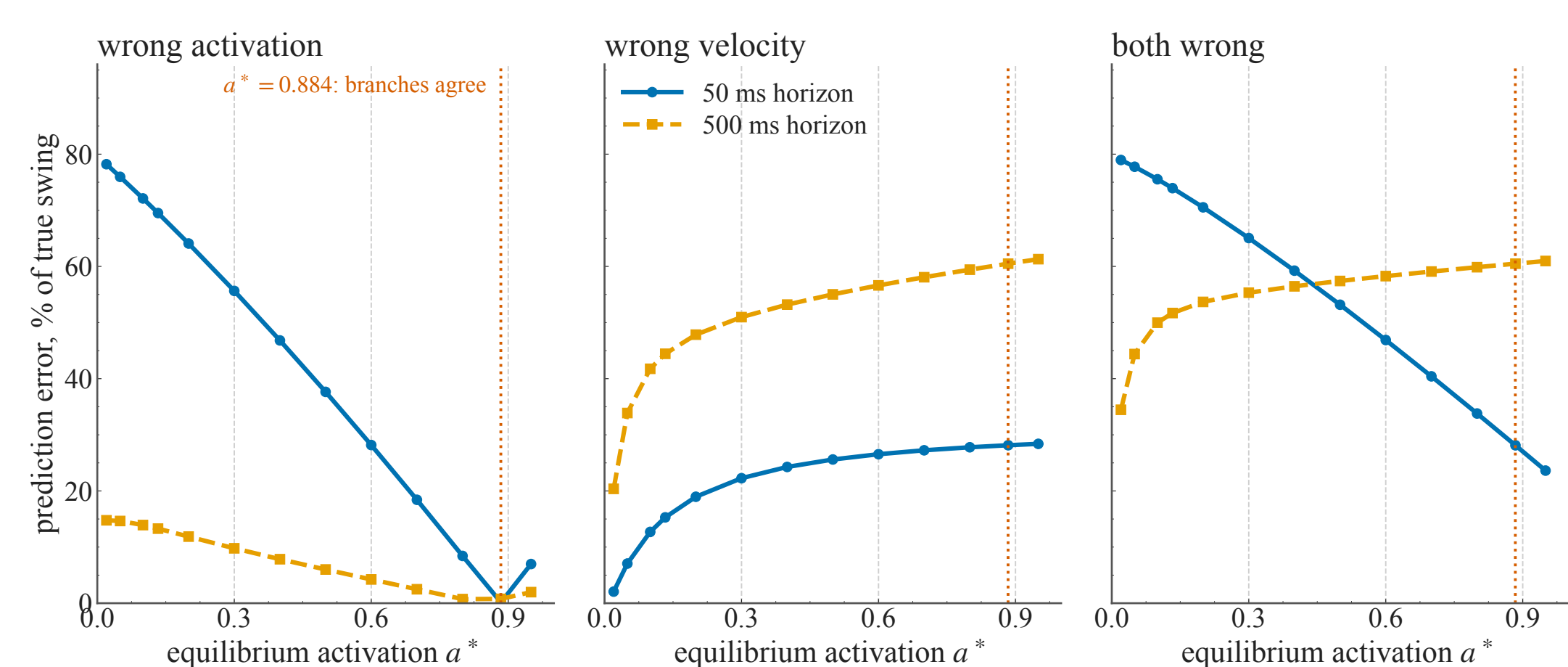


Figure 3. Held-load sweep at 60°: activation-branch error falls as the velocity-branch error grows. The combined error never vanishes in the sweep.

7 Results: what survives feedback

MPC uses identical weights, $N = 20$ and $T_s = 10$ ms. Blending the branches gives **8–9× more raising overshoot** than the naive, sign-test and relinearizing rules. Updating the model cuts lowering overshoot from **0.37° to 0.02°**.

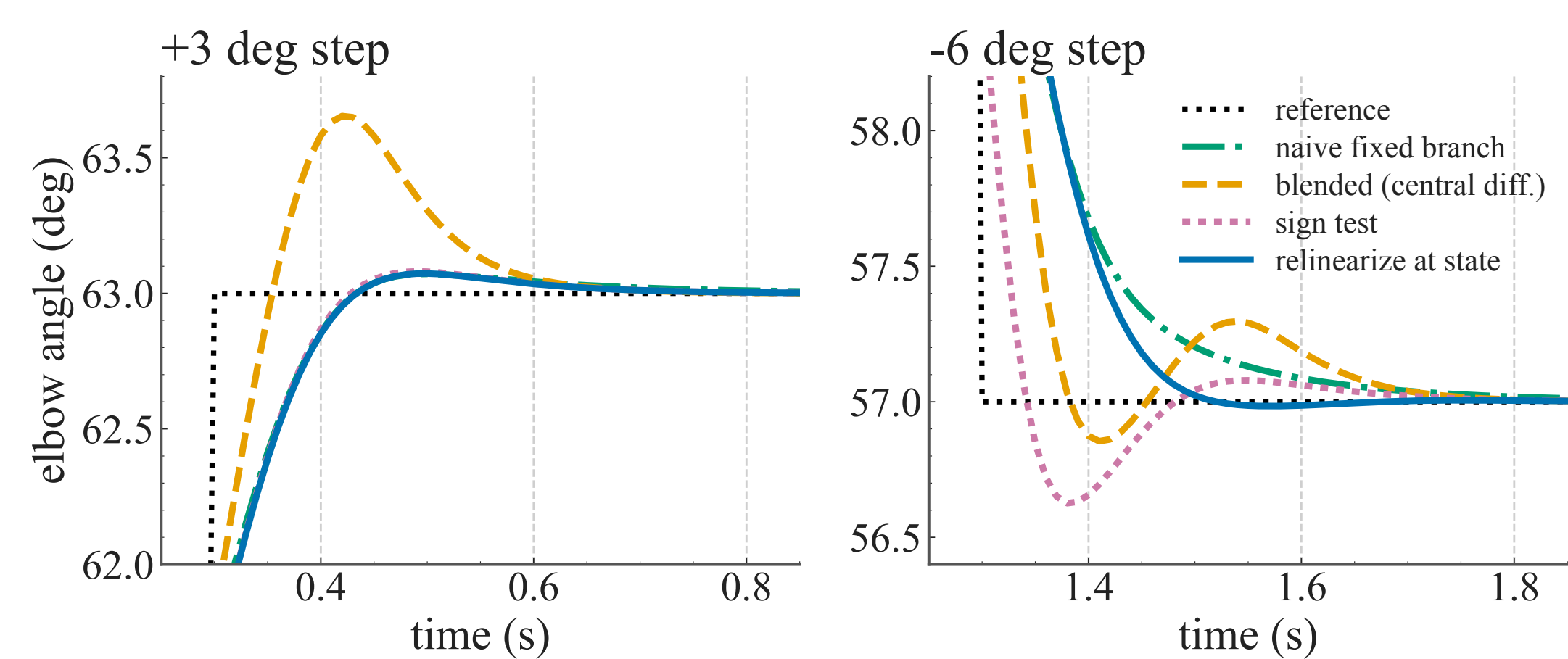


Figure 4. Tracking 60 → 63 → 57°. Branch selection affects transients even when delay-free feedback preserves stability and steady-state tracking.

8 Results: antagonist co-contraction

Adding triceps creates **three switches and eight modes**. Co-contraction can balance the muscles' directional damping and cancel the joint velocity kink; the cancellation depends on the moment arms.

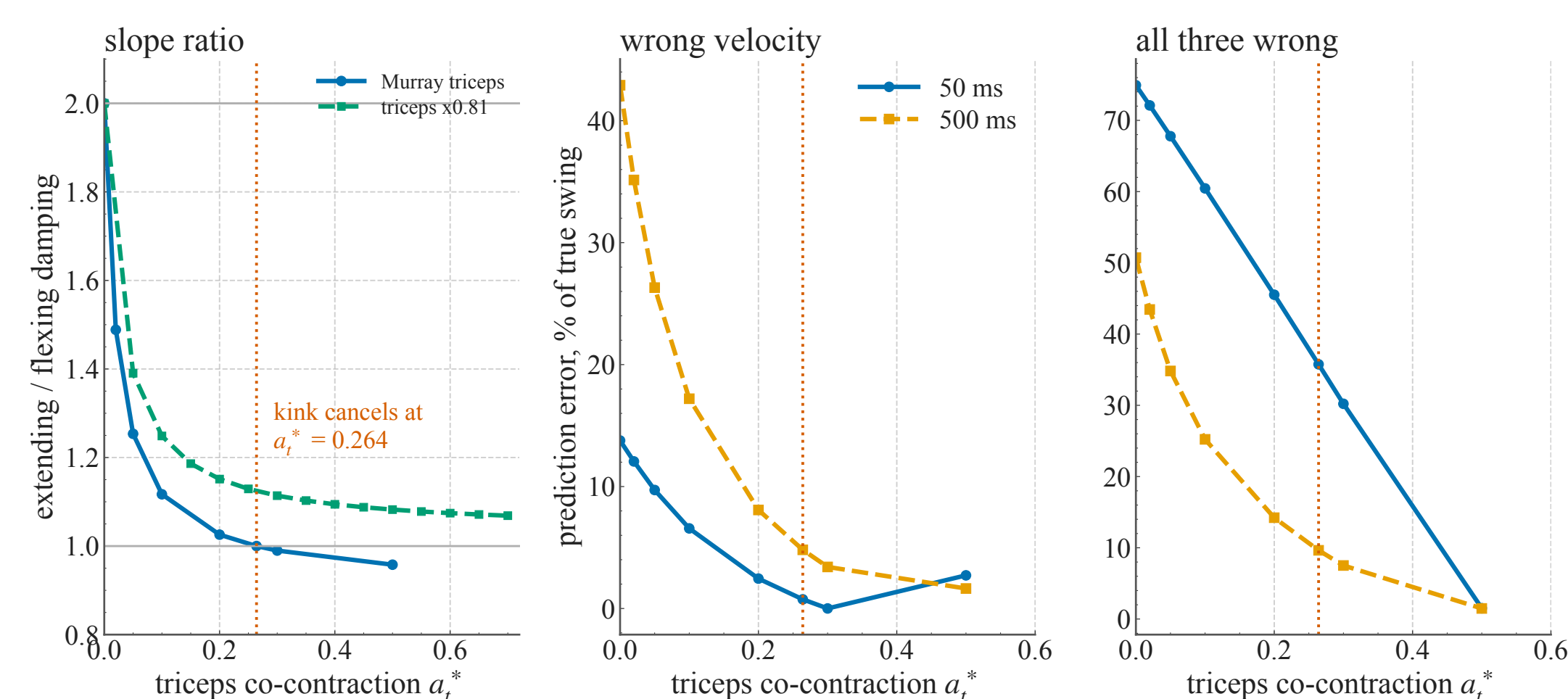


Figure 5. ICRA antagonist study at 60°. Co-contraction reduces the velocity slope ratio toward one and reduces wrong-branch errors, at higher effort.